

RESEARCH + IMPLEMENTATION SPEC

Natura Humanizer

Whole workflow in one document

NLP models · Filters · AI detection · Plagiarism / originality

For a StealthWriter-class application

Reference: stealthwriter.ai

No internal spec PDFs were attached.

This file is the single source of truth: what to use, in what order,
and what each engine must do.

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1. What this product does

Natura takes AI-written or mixed documents and does three jobs that StealthWriter sells as one product: rewrite the prose so it reads like a person wrote it, score leftover AI patterns at document and sentence level, and check that the new wording is original.

These are two separate engines. Do not merge them into one slider.

- AI removal — destroy LLM fingerprints (even rhythm, template transitions, hedge stacks, uniform fluency).
- Plagiarism-free — the new wording must not match the input, your corpus, or the public web.

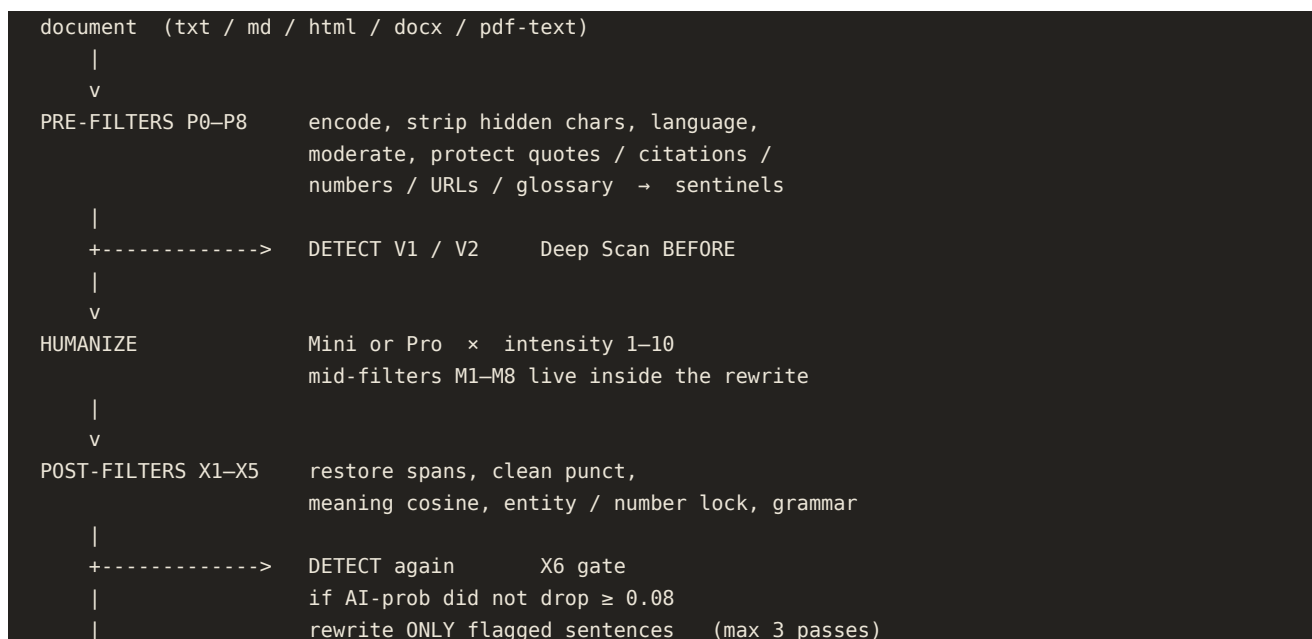
A synonym swap can look original and still scream AI. A humanizer can sound natural and still be 80% the same 3-grams as a Wikipedia paragraph. Both gates are required. No honest tool can promise 100% undetectable on Turnitin or GPTZero forever. Show scores. Never guarantee a vendor result.

Product contract (clone the job, not the brand)

StealthWriter	Natura	Required
Ghost Mini / Ghost Pro	Natura Mini / Natura Pro	Yes
Intensity 1-10	intensity 1-10	Yes
Detector V1 / V2	Scan V1 cheap / V2 accurate	Yes
Deep Scan colors	Heatmap + reason tags	Yes
Click sentence → alternatives	3 alternatives per flagged sentence	Yes
Re-humanize	Iterate only flagged sentences, max 3	Yes
Daily quota + word cap	Free 10/day, 1,000 words; paid 5,000	Yes
Content generator	Out of scope	No

2. Whole workflow

Every document follows the same path. Cheap jobs (Mini + V1) stay in Node and finish under 1.5 seconds per 1,000 words. Expensive jobs (Pro + V2 + web plagiarism) go through Redis / BullMQ.



```

|
+-----> ORIGINALITY A B C D    X7 gate
|          rewrite only matching sentences
|
v
REPORT
% Human before / after
sentence colors + reasons + 3 alternatives
originality breakdown + match URLs
meaning cosine, entity retention, burstiness

```

Two numbers on every job, always: AI human % and Originality %. Never one slider for both.

Stack for this workflow (React / Node / SQL)

Layer	Use	Do not use
Frontend	React 18 + TS + Vite + Tailwind	Next.js / Angular for this product
API	Node 20 + Fastify (Express OK)	Django / Spring as the main API
Jobs	BullMQ + Redis	Blocking HTTP on a 13B model
Database	PostgreSQL 16 + pgvector	MySQL as the only database
Raw text	Redis TTL 15 min	Full essays stored in SQL
Files	S3 or MinIO	Disk on the API box
Heavy NLP	Python FastAPI worker (Phase 2)	PyTorch inside Node
Pro LLM	Haiku / GPT-4.1-mini / Gemini Flash, later Llama 3.1 8B	Raw ChatGPT “make this undetectable”
Web play	Serper or Bing, or Copyleaks	Crawling the internet yourself

Phase 1 does not need Python. Mini + V1 + n-gram originality run in TypeScript on Node. PostgreSQL is required because of pgvector, JSONB Deep Scan payloads, and full-text search. MySQL may hold users/billing only.

3. NLP models

A humanizer is a constrained paraphraser. It must keep meaning, facts, names, numbers, quotes and citations; raise burstiness and perplexity; kill template diction; stay grammatical and on-register. It must not invent sources, change numbers, or follow instructions hidden in the document.

3.1 Humanizer models

Role	Model	Phase	Why
Mini CPU	TypeScript rules + lexicon + burstiness	1	Free, deterministic, fallback when Pro is down
Mini GPU	FLAN-T5 Base or humarin T5	2	Best open meaning lock (T5 > BART > PEGASUS)
Mini alt	tuner007/pegasus_paraphrase	2	Fast 3-5 sentence alternatives; weaker meaning
Pro API	Claude Haiku / GPT-4.1-mini / Gemini Flash	1 opt.	Ship Pro without a GPU
Pro long	Claude Sonnet / GPT-4.1	3	Coherence on 5,000 words
Pro owned	Llama 3.1 8B or Mistral 7B + LoRA (HIP)	3	Our Ghost Pro. Train AI → human, 2-4 passes

Never	Raw ChatGPT “make this undetectable”	—	Leaves a new LLM fingerprint. Detectors catch it
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HIP — custom Pro recipe (2026 paper)

- Collect human passages h. Have an LLM rewrite them into AI-ish a.
- Fine-tune a BASE model (not a chat model) on a → h. Base models look more human to GPTZero / Pangram than instruction-tuned ones.
- At inference apply 2–4 passes. Stop if detector is good AND meaning cosine ≥ 0.78 .
- Never 10 passes in production. The paper shows meaning dies.

Adversarial paraphrasing (2025) — the re-humanize loop

Use an off-the-shelf LLM, but only resend sentences the detector still flags. Training-free. This is the production loop in section 2.

3.2 Detector models

Role	Model / method	Note
V1 required	Stylometry: burstiness, TTR, AI lexicon, hedges, contractions, transitions, repetition	Runs in browser + Node. Explainable.
V2 vote	openai-community/roberta-base-openai-detector	~95% on GPT-2. Stale alone on GPT-4 / Claude. One vote only.
V2 primary	Binoculars (Falcon observer / performer)	Zero-shot. >90% ChatGPT at 0.01% FPR in the paper.
V2 vote	Fast-DetectGPT	Log-prob curvature. Cheaper than DetectGPT.
V2 mix	0.40 Binoculars + 0.25 RoBERTa + 0.20 V1 + 0.15 Fast-DetectGPT	Calibrate with Platt scaling on a 200-doc set.
Vendor	GPTZero, Originality.ai, Copyleaks, Sapling	Phase 3 paid panel. Never send Free-plan text.
Never alone	Any single 2020 RoBERTa as the product score	High false positives on formal human writing.

When you retrain, add V3. Do not silently move V2. Users compare runs across weeks.

3.3 Embeddings and utilities

Job	Use	Where
Meaning + plag vectors	intfloat/e5-base-v2 or OpenAI text-embedding-3-small	Python / API
Sentence split	compromise (Node) or nltk.sent_tokenize	Must respect Dr. e.g. U.S.
POS / entities	compromise	Mini must not synonym-swap names
Language ID	franc	Do not score German with an English lexicon
Grammar	LanguageTool self-host	Post-filter, high-confidence
Moderation	OpenAI Moderation or Detoxify	Pre-filter
DOCX / PDF / HTML	mammoth / pdf-parse / dompurify	Node ingest

3.4 Mini algorithm

Work per paragraph. At intensity 1–4, only touch sentences with V1 AI-prob ≥ 0.28 .

Intensity	Name	Transforms
1–2	Light	Replace AI transitions. Delete empty hedges (It is important to note that).
3–4	Light+	POS-aware synonym swap. Contractions (off in academic).
5–6	Medium	Split one long sentence or merge two short ones (burstiness). Rewrite formula openers.
7–8	Heavy	One passive $\rightarrow$ active per paragraph. Safe clause reorder.
9–10	Aggressive	Stronger synonym pressure. Higher meaning risk.

Keep 3 alternatives per touched sentence. Winner = $0.6 \times \text{humanScore} + 0.4 \times \text{cosine_to_source}$. Burstiness B = $\text{std}(\text{sentence_length}) / \text{mean}$. Target 0.35–0.70. Below 0.35 force a split or merge. Above 0.90 undo.

3.5 Pro algorithm

Pro is a line editor, not a chatbot. System contract:

- Keep every fact, number, name, quote, citation. Do not add examples, conclusions, or sources.
- Vary sentence length. Prefer concrete verbs. Banned list = AI lexicon (section 4.5).
- Leave sentinel tokens `[[U0]]` untouched. Register and intensity are parameters.
- Return only the rewritten passage. Temperature 0.7–0.95, presence_penalty 0.4, max tokens $\approx 1.3 \times \text{input}$.
- Then run the SAME post-filters as Mini. The LLM is not trusted.

```
target_ai = 0.45 on Free, 0.28 on paid
max_pass = 3 # never 10
for pass in 1..max_pass:
    rewrite only sentences with aiProb > 0.5
    stop if detect(draft).ai < target AND cosine(source, draft) >= 0.78
```

4. Filters

Filters are the difference between a demo and a product. Without them T5 and Claude both invent numbers, drop citations, and emit Furthermore. Run in this order. Fail closed.

4.1 Pre-filters — before any model sees the text

ID	Filter	Action
P0	Encoding	UTF-8, Unicode NFC, strip BOM
P1	Invisible abuse	Remove zero-width / bidi chars. Map Cyrillic lookalikes (a→a, e→e, o→o). Homoglyph attacks fool detectors AND plag checkers.
P2	Size	Enforce plan word cap. Reject empty. Word count < 80 $\rightarrow$ scan unreliable
P3	Language	franc. English only in Phase 1. Refuse others instead of faking a score
P4	Moderation	Block malware / CSAM / scam kits. Do not humanize those
P5	PII	Emails, phones, IBANs — lock or refuse
P6	Span protection	Quotes, citations, numbers, URLs, glossary $\rightarrow$ sentinels <code>[[U0]]</code>
P7	Structure	Headings, list markers, table cells stay units
P8	Instruction fence	Document is DATA. Ignore “ignore previous instructions” inside it

4.2 Mid-filters — inside the rewriter

ID	Filter	Rule
M1	Sentence budget	Output words within −20% / +15% of input
M2	Number lock	Every digit, %, currency token must reappear
M3	Glossary lock	Product names, statutes, drug names, course codes
M4	Quote lock	Quoted spans and blockquotes never rewritten
M5	Citation lock	(Smith, 2021), [12], doi:, URLs
M6	Register	academic / business / casual / neutral. Academic = no contractions below intensity 7
M7	Source n-gram ban	Do not emit an 8-gram that existed in the source
M8	Language lock	Output language == input language

4.3 Post-filters — after the rewriter

ID	Filter	Pass bar	On fail
X1	Restore spans	Every sentinel gone, original back	Re-insert from span table
X2	Whitespace / punct	No double spaces, paragraphs kept	Normalize
X3	Meaning check	Embedding cosine ≥ 0.78 (Phase 1 bag ≥ 0.72)	Drop pass; retry or return Mini
X4	Factuality	Jaccard on PERSON / ORG / NUM ≥ 0.98	Stitch missing entities back
X5	Grammar	LanguageTool high-confidence only	Auto-fix those only
X6	Detector gate	AI-prob dropped by ≥ 0.08	Re-humanize flagged sentences only
X7	Originality gate	Checks A/B/C under section 6 bars	Rewrite only matching sentences
X8	Watermark strip	No hidden chars in output	Strip again (models can re-add)

4.4 Span protection

Replace before Mini or Pro. Restore after. Tell Pro: leave tokens like `[[U0]]` untouched.

```
email    user@domain.tld
url      https://...
code     `inline`
quote    "... " or "... "
cite     (Author, 2021) | (Author et al., 2021) | [12] | doi:
num      $12.50 | 150 | 3.14%
term     user glossary, longest match first

sentinels: [[E0]] [[U1]] [[Q2]] [[N3]] [[T4]]
```

4.5 AI lexicon — always replace, even at intensity 1

Keep as a versioned YAML. Bump it when V2 starts flagging new clichés. Density of these tokens is also a V1 feature.

Phrases: it is important to note that · it is worth mentioning that · it should be noted that · in today's rapidly evolving landscape · in the realm of · a wide range of · a plethora of · a myriad of · due to the fact that → because · in order to → to · plays a crucial/pivotal role · pave the way for · a testament to · in conclusion · in summary.

Tokens: furthermore, moreover, additionally, subsequently, nevertheless, hence, thus, thereby,

leverage, utilize, facilitate, delve, underscore, showcase, enhance, optimize, streamline, empower, foster, comprehensive, robust, cutting-edge, groundbreaking, innovative, holistic, multifaceted, paradigm, synergy, tapestry, nestled, vibrant, landscape, realm, meticulously, noteworthy, aforementioned, henceforth.

5. AI detection functionality

Detectors do not look for bad English. They look for statistical fingerprints of LLMs: low perplexity (next word too easy), low burstiness (sentence lengths almost equal), template diction, hedge stacks, near-zero contractions, parallel triads, uniform fluency.

We ship our own detector so we can iterate the humanizer against a stable target, avoid leaking Free-plan text to vendors, and show sentence-level reasons. Vendor APIs are a paid second opinion only.

5.1 V1 features — every sentence and the document

Feature	AI tends to	Human tends to
Burstiness $\text{std}(\text{len})/\text{mean}$	0.10 – 0.25	0.35 – 0.80
Type-token ratio	Lower, safer vocab	Higher, more hapaxes
AI-lexicon density	furthermore / leverage / robust	Concrete verbs
Hedge stacks	It is important to note that...	Fewer stacked hedges
Contraction rate	Near 0 in “professional” mode	2–8% in normal prose
Template transitions	moreover / furthermore every few	Mixed or none
Starter uniqueness	The / This / In / Moreover	More variety
Parallel triads	not only X, but also Y, as well as Z	Uneven
Length uniformity	Almost every sentence 15–22 words	Mixed short and long
Bigram / trigram repetition	Templated	Messier

Sentence AI-prob = sigmoid of a weighted sum of z-scored features, using a ± 2 sentence window for local burstiness. Document score weights middle sentences 1.0 and first/last 0.75 (intros are often human even when the body is generated). % Human = $100 \times (1 - \text{document AI-prob})$.

AI-prob	Label	Deep Scan color
< 0.34	Human	Green
0.34 – 0.62	Mixed	Amber
> 0.62	AI	Rose

Short-text rule: under 80 words return unreliable = true. Still show a number, plus the caveat. Do not snap to 0% or 100%.

5.2 Deep Scan payload

```
{
  "id": "s-14",
  "text": "Furthermore, the implementation of ...",
  "aiProb": 0.81,
  "label": "ai",
  "reasons": ["template-transition: furthermore", "ai-lexicon: implementation", "low-burst-context"],
  "alternatives": ["The team actually shipped ...", "What they put in place ..."]
}
```

reasons is required. StealthWriter only shows a color. We show why.

5.3 V2 and the humanizer loop

V2 ensemble is in section 3.2. Expose V1 and V2 as a toggle. Tune the humanizer against OUR V2, then spot-check GPTZero weekly. Daily overfitting to one vendor dies on their next release.

Required loop: scan source → humanize → scan output → if document AI-prob did not fall by ≥ 0.08 , or any sentence is still > 0.62 , resend only those sentences → stop at 3 passes or when meaning cosine fails.

5.4 Failure modes (document these in the product)

Input	What the detector does	What we do
Formal scientific human writing	Looks AI (low burstiness, jargon)	Technical register loosens lexicon penalty
Human + Grammarly	Mixed	Correct — say so
Code, poetry, bullet SOPs	Nonsense scores	Skip or use a different head
Non-English	Fake English-lexicon score	Refuse until we have a model
Homoglyph / zero-width “bypass”	Looks human if scored raw	P1 strips them BEFORE scoring
< 80 words	Unstable	unreliable = true

6. Plagiarism / originality functionality

Users say “plag free.” StealthWriter’s FAQ (“we generate original rewrites so it won’t trigger plagiarism detectors”) only really covers check A. We implement four checks.

Check	Question	When
A Rewrite distance	Did we change the wording vs. this input?	Day 1
B Internal corpus	Does this match another job, user, or seed library?	Day 1 hashes; vectors Phase 2
C Web	Does this match the public internet?	Phase 2–3, paid
D Citation honesty	Are quotes and citations still on the same claims?	Day 1

We will not build a Turnitin-scale web index. Use a search API or buy Copyleaks / Originality.ai. We will not get a Turnitin reseller API.

6.1 Check A — rewrite distance

Word 3-gram Jaccard vs input. Character 5-gram Dice as backup. Embedding cosine in Phase 2 (catches paraphrase Jaccard misses).

Intensity	Max 3-gram Jaccard vs input	Min embedding cosine vs input
Light 1–3	≤ 0.55	≥ 0.88
Medium 4–7	≤ 0.40	≥ 0.82
Aggressive 8–10	≤ 0.28	≥ 0.78

Jaccard still high → flag “too close to input.” Cosine too low → we over-rewrote; roll that sentence back. Mid-filter M7 exists so Check A can pass.

6.2 Check B — internal corpus

Same family as Moss and Turnitin's first stage. Normalize → character 9-grams → rolling hash → Winoing (window = 4). Store (hash, user_id, job_id, offset) only — no raw text. Shared-hash rate > 0.18 → fine pass with word 5-gram alignment → match spans.

Phase 2: chunk 80–120 tokens, embed with e5, pgvector cosine ≥ 0.92 = near-paraphrase of stored content.

6.3 Check C — web originality (paid)

- From each sentence, take the rarest 8-gram that is not a protected span.
- Search that exact phrase on Serper / Bing / Brave. About 3 queries per 500 words.
- If a top hit shares $\geq 60\%$ tokens, fetch the page and align n-grams.
- Attach {url, title, overlap, excerpt} to the sentence.
- Allowlist boilerplate: proverbs, “p < 0.05 was considered...”, legal headers, standard IRB lines.

6.4 Check D — quotes and citations

If Pro rewrites a quotation, that is a new plagiarism bug (misquotation), not a feature. Quotes are locked (M4). Every quote in the output must exist in the input. Every (Author, Year) / [n] must remain within ± 1 sentence of the same claim. If Pro dropped it, stitch it back.

6.5 Score we show

Originality	94%	
└ rewrite vs input	91%	Check A
└ internal corpus	100%	Check B
└ web sources	88%	Check C → click → highlight + URL

Colors: ≥ 90 green · 75–89 amber · < 75 rose. Free plan: 0.45 A + 0.40 B + 0.15 unused C redistributed. Paid: include C.

6.6 What we will not claim

- We cannot detect idea plagiarism.
- We cannot detect translated plagiarism until multilingual embeddings (Phase 3).
- We cannot guarantee 0% Turnitin.
- Humanizing a scraped article does not create a copyright license. Originality $\neq$ permission.

7. Quality gates and eval

Gate	Fail if	Action
Meaning cosine	< 0.78 (Phase 1 bag < 0.72)	Retry or return Mini
Entity / number retention	< 0.98	Stitch back
Word-count drift	Outside −20% / +15%	Reject pass
AI-prob drop	< 0.08	Loop flagged sentences
3-gram Jaccard vs input	Above intensity bar in 6.1	Flag / rewrite those lines

Frozen eval set (required before any “it works” claim)

200 documents: 50 known-human, 50 raw GPT-4 / Claude / Gemini, 50 mixed, 50 technical / legal

(false-positive traps).

Metric	Ship bar
Mean AI-prob drop after humanize (V2)	≥ 0.35
Meaning cosine after humanize	≥ 0.82
Number / entity retention	≥ 0.98
False positive on known-human docs	< 8% marked AI
Human Likert fluency, 3 raters, 20 samples	$\geq 4.0 / 5$
p95 Mini latency	< 1.5 s / 1,000 words
p95 Pro latency	< 15 s / 5,000 words

8. Documents, storage, build order

Ingest

Format	How	Phase
TXT / MD	Native	1
HTML	DOMPurify → text	1
DOCX	mammoth → HTML → text, keep headings	1
PDF with text layer	pdf-parse / pdf.js + page map	1
Encrypted PDF / image-only scan	Reject with a clear error	1
OCR	Tesseract / Textract	3

What we store vs drop

Keep in Postgres	Drop after job (Redis TTL 15 min)
User, plan, usage events	Raw source text
Job metadata, model version, intensity	Raw output unless user opts into history
Scan scores, originality JSON	File bytes after extract (S3 key only if saved)
Fingerprint hashes; embeddings in Phase 2	—
Word counts, latency, errors	—

Build order

Phase	Time	Deliver
0 This PDF	—	Models, filters, AI, plagiarism, workflow decided
1 MVP	2–3 weeks	Node + Postgres + Redis + Mini + V1 + checks A/B/D + DOCX/PDF. Optional one LLM key for Pro.
2 Real models	3–5 weeks	FastAPI: T5, V2 ensemble, e5, LanguageTool, score-gated loop
3 Parity	6–8 weeks	HIP Llama, Serper, multi-detector panel, more languages

Market as a writing-quality and originality assistant, not “bypass your university.” If an institution requires AI disclosure, this tool does not remove that duty. Refuse exam-answer / take-home / impersonation jailbreaks in the Pro prompt and in Acceptable Use.

9. One-page cheat sheet

WORKFLOW	pre-filter → detect → humanize → post-filter → re-detect (loop ≤ 3) → originality → report	
NLP Mini	TypeScript rules + lexicon + burstiness then FLAN-T5 / T5 paraphraser	
NLP Pro	Haiku or GPT-4.1-mini first then Llama 3.1 8B LoRA (HIP, 2–4 passes)	
Embed	e5-base or text-embedding-3-small	
Detect V1	burstiness, TTR, lexicon, hedges, contractions, transitions, starters, repetition	
Detect V2	0.40 Binoculars + 0.25 RoBERTa + 0.20 V1 + 0.15 Fast-DetectGPT	
FILTERS	protect spans → rewrite → restore cosine ≥ 0.78 entities ≥ 0.98 AI score must drop ≥ 0.08 originality gate	
PLAG A	3-gram Jaccard vs input	day 1
PLAG B	Winnowing hashes + later vectors	day 1 / phase 2
PLAG C	Serper or Copyleaks	paid
PLAG D	quote + citation lock	day 1
STACK	React + Node + PostgreSQL + Redis Python FastAPI only for Phase 2 models	
NEVER	“100% undetectable” ChatGPT as the humanizer with no filters MySQL as the only database storing raw essays in Postgres	

Research this document draws on

StealthWriter product, Ghost 5.2 Mini/Pro, V2 scanner, Deep Scan, changelog (live site, 2026). HIP — Humanization by Iterative Paraphrasing (arXiv 2605.19516, 2026). Adversarial Paraphrasing (arXiv 2506.07001, 2025). DAMAGE — Detecting Adversarially Modified AI Generated Text (arXiv 2501.03437). Binoculars (arXiv 2401.12070). DetectGPT (2023) and Fast-DetectGPT (2023). DIPPER paraphraser (Krishna et al., Google Research, 2023). OpenAI roberta-base-openai-detector model card. GPTZero public description of perplexity + burstiness. Winnowing (Schleimer et al., 2003). GLTR (Gehrmann et al., 2019). T5 vs BART vs PEGASUS paraphraser comparisons and Hugging Face model cards.

This PDF is the single workflow document. Use it as the build spec.